Titanic Data Workflow and Interpretation

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AI Mastery Capstone

AI Programming Foundations

October 30, 2026

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Overview

I built a Python notebook that loads, cleans, explores, and visualizes the Titanic passenger dataset distributed in the Seaborn example repository (Waskom, n.d.). The source CSV is available at https://github.com/mwaskom/seaborn-data/blob/master/titanic.csv. The workflow preserves all 891 passenger rows and compares recorded ages and survival percentages by sex and passenger class.

Dataset Description

The CSV contains 891 rows and 15 columns describing passenger characteristics and survival outcomes. The main variables are survived, pclass, sex, age, and fare; survived uses 1 for survival and 0 for non-survival. Other fields describe family relationships, embarkation, and deck information. Ages are recorded for 714 passengers, leaving 177 missing values (19.9%), while deck has 688 missing values (77.2%). Both embarked and embark_town have two missing values, and the retained survival and passenger-class fields have no missing values.

Workflow Description

Ingestion uses pandas.read_csv("titanic.csv"), displays the first five rows, and checks the minimum dimensions. Inspection then reports data types, missing-value counts and percentages, and the number of identical rows. Two cleaning functions create a working copy, remove six unused derived or overlapping columns, and label missing embarkation ports as Unknown. The EDA function returns numeric summaries and grouped passenger counts, survivor counts, and survival percentages. Matplotlib displays and saves three labeled plots. The final Markdown section interprets the results.

Danchev (2022) describes an approach that connects Python code, narrative, reproducible workflow, and critical interpretation in Jupyter notebooks. I applied that approach by keeping the loading, cleaning, summaries, plots, and interpretation together in one ordered notebook rather than performing undocumented spreadsheet edits.

Key Decisions and Assumptions

The function remove_redundant_columns removes class, who, adult_male, embark_town, alive, and alone. These fields overlap with retained information or provide derived groupings that are not required for the selected comparisons. For example, pclass retains the class codes, embarked retains the embarkation codes, and survived retains the outcome. Removing these columns is a scope and readability choice; it does not mean that their values are invalid. The original dataframe remains unchanged in df, and the working dataset contains 891 rows and nine columns.

The function label_missing_embarkation replaces two missing embarked values with Unknown. This explicitly records unavailable information rather than assigning a likely port. The label is used for descriptive data handling, not as a validated statistical solution to missing predictor values. I retained missing ages and deck values and did not delete passengers with incomplete records. Sterne et al. (2009) explain that missing-data handling can introduce bias and that reporting missingness and the associated assumptions is important. I therefore report the 177 excluded ages alongside Figure 1.

Identical rows are retained because this dataset does not include a unique passenger identifier, and matching characteristics alone do not establish that two records describe the same person. Fare values of zero and unusually high fares are also retained because the notebook provides no evidence that these values are recording errors. The EDA focuses on directly

interpretable group summaries rather than prediction. Figure 1 uses 20 bins to show the shape of recorded ages; Figures 2 and 3 use percentage scales from 0 to 100 to compare survival across groups of different sizes.

Results and Interpretation

Figure 1 shows that recorded ages are concentrated in young adulthood, with fewer passengers at older ages and some young children. The mean recorded age is 29.7 years, the median is 28 years, and the range is 0.42 to 80 years. These statistics describe the recorded ages only.

Figure 2 shows that 233 of 314 female passengers survived (74.2%), compared with 109 of 577 male passengers (18.9%). This is a substantial observed difference, but the notebook does not adjust for age, class, or other characteristics.

Figure 3 shows survival of 136 of 216 first-class passengers (63.0%), 87 of 184 second-class passengers (47.3%), and 119 of 491 third-class passengers (24.2%). Survival decreases across the three class categories in this dataset. These unadjusted comparisons do not establish causation. The fare summary also shows a mean of 32.20, a median of 14.45, and a maximum of 512.33, suggesting a right-skewed distribution; fare is summarized numerically rather than shown in a separate figure.

Responsible Practice

The age histogram could be misleading if passengers with missing ages differ systematically from those with recorded ages. Keeping all rows protects the sex and class summaries from unnecessary age-based exclusions, but it does not remove potential bias in the age histogram. Sterne et al. (2009) recommend examining differences between complete and incomplete records and clearly reporting exclusions. My next check would compare age-

missingness rates across sex, passenger class, and survival status before choosing an imputation approach or extending age-based analysis.

Deck is too incomplete for the current analysis to support a detailed description of all passengers by deck, so it is not used in the plots. Unknown embarkation is an explicit missing-information label and should not be interpreted as a real port. The dataset contains recorded binary sex categories; this analysis reports those historical labels without making claims about groups that are not represented. The survival patterns are specific to this dataset and are not a basis for decisions about present-day individuals. Any future automated cleaning should preserve the source file and record the proposed changes for review.

Reproducibility

The repository is https://github.com/perrym/ai-programming-foundations-project. The Git workflow uses main and development, with changes recorded through task-based commits before integration into main. The notebook reads the local CSV through a relative path, and no network connection or random sampling is needed for the analysis once the files and packages are available. Keeping code, explanations, and outputs together follows the reproducible workflow emphasis in Danchev (2022).

To reproduce the work, clone the repository, create a dedicated Python environment, and install the dependencies with pip install -r requirements.txt. Generate requirements.txt in the project environment using pip freeze > requirements.txt so that it records the installed package versions. Open data_workflow.ipynb with python -m notebook, restart the kernel, and run all cells in order. Keep titanic.csv beside the notebook; the three figure files are written to that same working directory. The uploaded notebook records Python 3.11.6, but complete environment matching also requires the dependency file.

Future Integration Reflections

For a machine learning extension, I would use survived as the target, split the data before fitting preprocessing, and compare a simple baseline with an interpretable classifier. I would explicitly exclude alive because it duplicates the target, review the other derived fields, and fit any imputation and encoding only on the training portion. Before training a neural network, I would prepare numeric inputs, encode categorical variables, check scaling and missing values, and assess whether 891 records justify that model compared with a simpler baseline. Agentic automation could help run validation checks, regenerate plots, and flag changed missing-value counts. It would be risky to let an agent delete identical rows, impute ages, or change interpretation without a recorded rationale and human review.

Summary and Interpretation

The workflow shows clear descriptive differences in survival by recorded sex and passenger class while retaining every passenger row. The age distribution is concentrated in young adulthood, but almost one fifth of ages are missing. Missing-data assumptions, overlapping fields, possible recording issues in fares, and unadjusted group comparisons limit the conclusions. The main outcome is an ordered, reusable analysis whose cleaning decisions and observations can be inspected alongside the code.

Figures

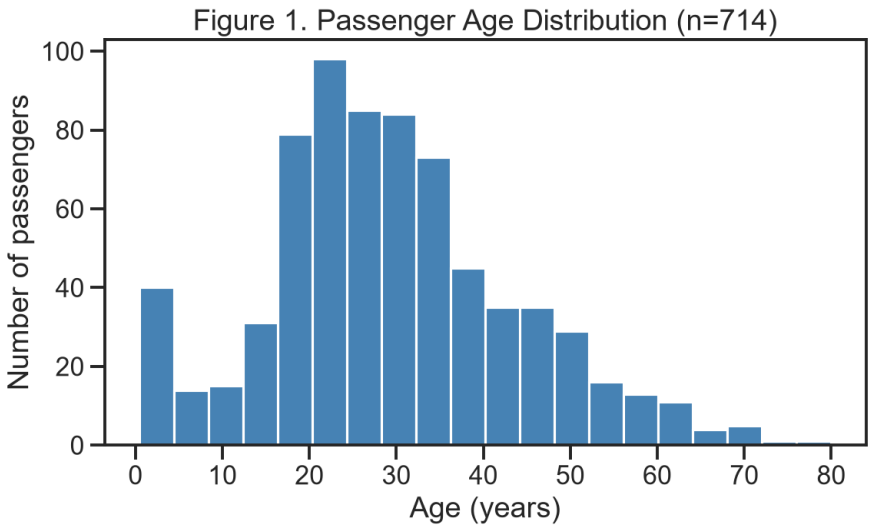


Figure 1. Recorded ages for 714 passengers. The 177 missing ages are excluded only from this histogram.

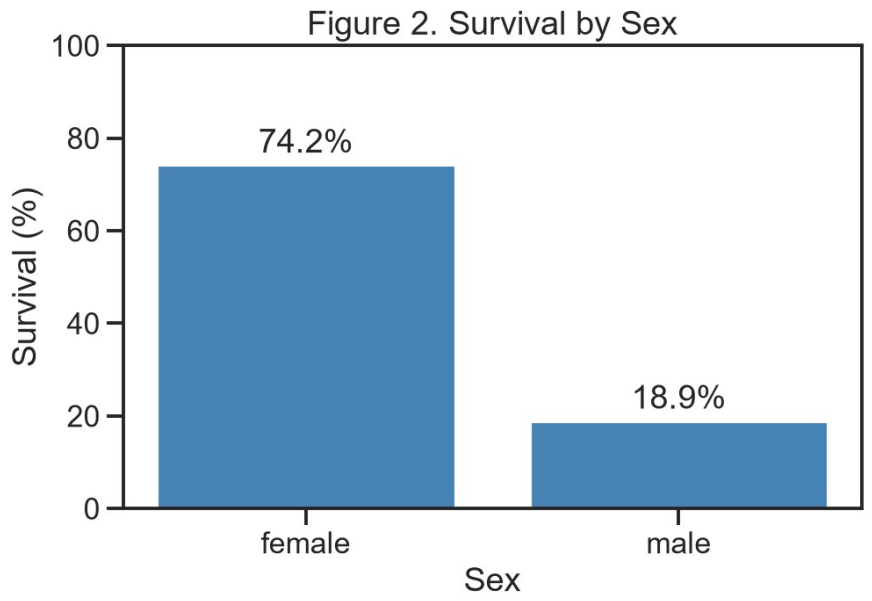


Figure 2. Observed survival percentages within each recorded sex category. Group sizes are 314 female and 577 male passengers.

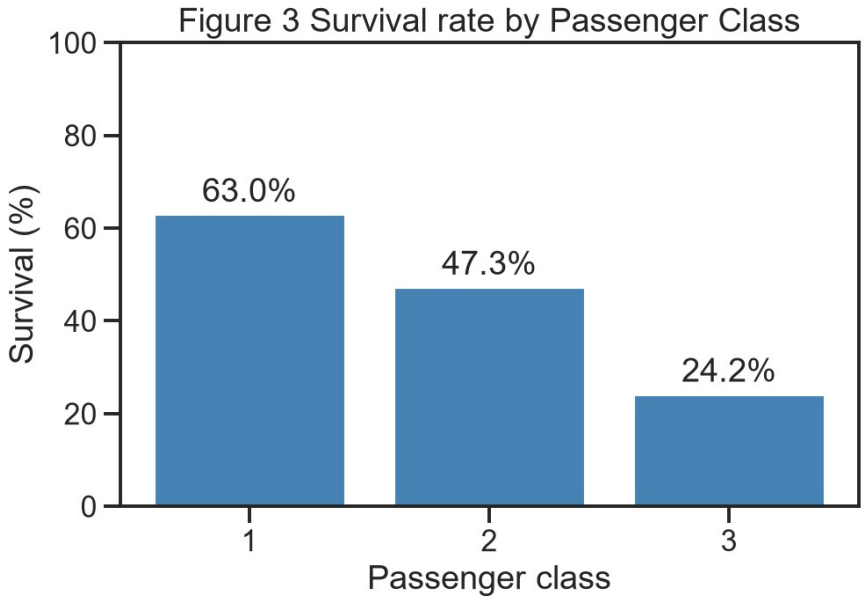


Figure 3. Observed survival percentages by passenger class. Group sizes are 216, 184, and 491 for classes 1, 2, and 3.

References

Danchev, V. (2022). Reproducible data science with Python: An open learning resource. *Journal of Open Source Education, 5*(56), Article 156. https://doi.org/10.21105/jose.00156

Sterne, J. A. C., White, I. R., Carlin, J. B., Spratt, M., Royston, P., Kenward, M. G., Wood, A. M., & Carpenter, J. R. (2009). Multiple imputation for missing data in epidemiological and clinical research: Potential and pitfalls. *BMJ, 338,* Article b2393. https://doi.org/10.1136/bmj.b2393

Waskom, M. (n.d.). *titanic.csv* [Data set]. GitHub. https://github.com/mwaskom/seaborn-data/blob/master/titanic.csv